

Source:

HTTP Request:

1. Frame:

It acts as a digital receipt for the captured packet the exact moment it arrives at your computer, which means it records the precise timestamp, physical size, and the specific network interface used, and ensures you have the exact background metadata needed to measure network delays and see the full structure of the traffic.

```
▼ Frame 15: 408 bytes on wire (3264 bits), 408 bytes captured (3264 bits) on interface \Device\NPF_{3545F9A5-3D9A-47AE-8939-48C3831E524D}
  Section number: 1
  ▼ Interface id: 0 (\Device\NPF_{3545F9A5-3D9A-47AE-8939-48C3831E524D})
    Interface name: \Device\NPF_{3545F9A5-3D9A-47AE-8939-48C3831E524D}
    Interface description: Ethernet 2
    Encapsulation type: Ethernet (1)
    Arrival Time: Feb 28, 2026 14:10:13.714604000 Bangladesh Standard Time
    UTC Arrival Time: Feb 28, 2026 08:10:13.714604000 UTC
    Epoch Arrival Time: 1772266213.714604000
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 0.920823000 seconds]
    [Time delta from previous displayed frame: 0.996813000 seconds]
    [Time since reference or first frame: 2.009894000 seconds]
    Frame Number: 15
    Frame Length: 408 bytes (3264 bits)
    Capture Length: 408 bytes (3264 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ip:tcp:http]
    [Coloring Rule Name: HTTP]
    [Coloring Rule String: http || tcp.port == 80 || http2]
```

2. Ethernet:

Source MAC Address: 1c:69:7a:f8:43:c9

Destination MAC Address: c0:f6:ec:90:a9:0b

It uses these MAC addresses to deliver data frames directly from one physical device to the very next connected device on the same local network, ensuring the data takes the correct physical "hop" on its journey. This section identifies the hardware MAC addresses and resolved device names for both the source and destination. It also indicates the encapsulated network protocol (e.g., IPv4) and concludes with a checksum to verify data integrity."

```
▼ Ethernet II, Src: EliteGroupCo_f8:43:c9 (1c:69:7a:f8:43:c9), Dst: HuaweiTechno_90:a9:0b (c0:f6:ec:90:a9:0b)
  > Destination: HuaweiTechno_90:a9:0b (c0:f6:ec:90:a9:0b)
  > Source: EliteGroupCo_f8:43:c9 (1c:69:7a:f8:43:c9)
  Type: IPv4 (0x0800)
```

3. Internet Protocol Version:

It handles the routing and delivery of packets across the network, ensuring they travel from the original source to the final destination. This section contains the logical IP addresses of both the sender and receiver, identifies the encapsulated protocol, and specifies the total size of the packet.

Internet Protocol Version 4, Src: 172.18.175.51, Dst: 103.230.106.216

```
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  Total Length: 394
  Identification: 0x10b0 (4272)
> 010. .... = Flags: 0x2, Don't fragment
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 128
  Protocol: TCP (6)
  Header Checksum: 0x0000 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 172.18.175.51
  Destination Address: 103.230.106.216
```

4. Transmission Control Protocol (TCP):

The Transmission Control Protocol (TCP) section represents the Transport Layer (Layer 4), acting as a strict, guaranteed delivery system for the data. The Transmission Control Protocol (TCP) is a reliable, connection-oriented protocol used to ensure that data is delivered accurately and in order between devices over a network. Before data transfer begins, TCP establishes a connection using a three-way handshake: first, the client sends a SYN packet to the server to request a connection; second, the server replies with a SYN-ACK packet to acknowledge the request and signal readiness; finally, the client sends an ACK packet back to confirm the connection. After this SYN → SYN-ACK → ACK process is completed, a reliable communication session is established and data transmission can begin.

Transmission Control Protocol, Src Port: 54288, Dst Port: 80, Seq: 709, Ack: 803, Len: 354

```
Source Port: 54288
Destination Port: 80
[Stream index: 0]
> [Conversation completeness: Incomplete (12)]
  [TCP Segment Len: 354]
  Sequence Number: 709 (relative sequence number)
  Sequence Number (raw): 2006540716
  [Next Sequence Number: 1063 (relative sequence number)]
  Acknowledgment Number: 803 (relative ack number)
  Acknowledgment number (raw): 1818678326
  0101 .... = Header Length: 20 bytes (5)
> Flags: 0x018 (PSH, ACK)
  Window: 64733
  [Calculated window size: 64733]
  [Window size scaling factor: -1 (unknown)]
  Checksum: 0x2f81 [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
> [Timestamps]
> [SEQ/ACK analysis]
  TCP payload (354 bytes)
```

5. Hypertext Transfer Protocol (HTTP):

Hypertext Transfer Protocol (HTTP) works in the Application Layer (Layer 7) of the OSI model. It is the protocol used for communication between a web browser (client) and a web server to request and transfer web pages, images, and other resources on the internet. HTTP sends its data through the Transmission Control Protocol (TCP) at the Transport Layer, which ensures reliable delivery of the requested information between devices. It acts as the direct request to the server, specifying exactly which website or application the user wants to access. It also bundles essential details like the specific files being requested, instructions to keep the connection open for faster loading, and other crucial metadata

```
✓ Hypertext Transfer Protocol
  > GET /time.php HTTP/1.1\r\n
    Host: cspb.teletalk.com.bd\r\n
    Connection: keep-alive\r\n
    User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/145.0.0.0 Safari/537.36\r\n
    Accept: */*\r\n
    X-Requested-With: XMLHttpRequest\r\n
    Referer: http://cspb.teletalk.com.bd/\r\n
    Accept-Encoding: gzip, deflate\r\n
    Accept-Language: en-US,en;q=0.9\r\n
    \r\n
    [Full request URI: http://cspb.teletalk.com.bd/time.php]
    [HTTP request 3/19]
    [Prev request in frame: 10]
    [Response in frame: 17]
    [Next request in frame: 26]
```

Destination:

HTTP Response:

1. Frame:

It acts as a tracking system, indicating whether this specific packet is a direct reply to an earlier request. It also records the exact size of the frame and its assigned sequence number within the capture.

```

▼ Frame 3: 455 bytes on wire (3640 bits), 455 bytes captured (3640 bits) on interface \Device\NPF_{3545F9A5-3D9A-47AE-8939-48C3831E524D}, id 0
  Section number: 1
  ▼ Interface id: 0 (\Device\NPF_{3545F9A5-3D9A-47AE-8939-48C3831E524D})
    Interface name: \Device\NPF_{3545F9A5-3D9A-47AE-8939-48C3831E524D}
    Interface description: Ethernet 2
    Encapsulation type: Ethernet (1)
    Arrival Time: Feb 28, 2026 14:10:11.707063000 Bangladesh Standard Time
    UTC Arrival Time: Feb 28, 2026 08:10:11.707063000 UTC
    Epoch Arrival Time: 1772266211.707063000
    [Time shift for this packet: 0.000000000 seconds]
    [Time delta from previous captured frame: 0.001224000 seconds]
    [Time delta from previous displayed frame: 0.002353000 seconds]
    [Time since reference or first frame: 0.002353000 seconds]
    Frame Number: 3
    Frame Length: 455 bytes (3640 bits)
    Capture Length: 455 bytes (3640 bits)
    [Frame is marked: False]
    [Frame is ignored: False]
    [Protocols in frame: eth:ethertype:ip:tcp:http:data-text-lines]
    [Coloring Rule Name: HTTP]
    [Coloring Rule String: http || tcp.port == 80 || http2]

```

2. Ethernet:

It guarantees that the transmitted data reaches the correct destination using MAC addresses and also checks for errors during the transmission process.

```

▼ Ethernet II, Src: HuaweiTechno_90:a9:0b (c0:f6:ec:90:a9:0b), Dst: EliteGroupCo_f8:43:c9 (1c:69:7a:f8:43:c9)
  > Destination: EliteGroupCo_f8:43:c9 (1c:69:7a:f8:43:c9)
  > Source: HuaweiTechno_90:a9:0b (c0:f6:ec:90:a9:0b)
  Type: IPv4 (0x0800)

```

3. Internet Protocol:

It is responsible for routing data packets correctly across a network. It contains the source and destination IP addresses as required and helps ensure that the information is delivered to the correct destination.

```

▼ Internet Protocol Version 4, Src: 103.230.106.216, Dst: 172.18.175.51
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
  Total Length: 441
  Identification: 0xb483 (46211)
  > 010. .... = Flags: 0x2, Don't fragment
  ...0 0000 0000 0000 = Fragment Offset: 0
  Time to Live: 249
  Protocol: TCP (6)
  Header Checksum: 0x9db6 [validation disabled]
  [Header checksum status: Unverified]
  Source Address: 103.230.106.216
  Destination Address: 172.18.175.51

```

4. Transmission Control Protocol:

It manages the delivery of data packets over the internet. It includes acknowledgement and segment numbers and helps ensure reliable transmission and proper flow control.

```
▼ Transmission Control Protocol, Src Port: 80, Dst Port: 54288, Seq: 1, Ack: 355, Len: 401
  Source Port: 80
  Destination Port: 54288
  [Stream index: 0]
  > [Conversation completeness: Incomplete (12)]
  [TCP Segment Len: 401]
  Sequence Number: 1 (relative sequence number)
  Sequence Number (raw): 1818677524
  [Next Sequence Number: 402 (relative sequence number)]
  Acknowledgment Number: 355 (relative ack number)
  Acknowledgment number (raw): 2006540362
  0101 .... = Header Length: 20 bytes (5)
  > Flags: 0x018 (PSH, ACK)
  Window: 65535
  [Calculated window size: 65535]
  [Window size scaling factor: -1 (unknown)]
  Checksum: 0xb10d [unverified]
  [Checksum Status: Unverified]
  Urgent Pointer: 0
  > [Timestamps]
  > [SEQ/ACK analysis]
  TCP payload (401 bytes)
```

5. Hypertext Transfer Protocol:

This protocol is used to communicate the server's response to a client's request. It tells the sender whether the request was successful or not. An HTTP response includes a status line, response headers, and a response body. The status line contains the HTTP version and status code, the headers provide additional information about the response, and the body contains the requested resource or data.

```
▼ Hypertext Transfer Protocol, has 2 chunks (including last chunk)
  > HTTP/1.1 200 OK\r\n
    Server: nginx\r\n
    Date: Sat, 28 Feb 2026 08:10:01 GMT\r\n
    Content-Type: text/html; charset=UTF-8\r\n
    Transfer-Encoding: chunked\r\n
    Connection: keep-alive\r\n
    Vary: Accept-Encoding\r\n
    X-Powered-By: PHP/5.6.40\r\n
    X-Frame-Options: SAMEORIGIN\r\n
    X-XSS-Protection: 1; mode=block\r\n
    Content-Encoding: gzip\r\n
    \r\n
    [HTTP response 1/19]
    [Time since request: 0.002353000 seconds]
    [Request in frame: 1]
    [Next request in frame: 10]
    [Next response in frame: 12]
    [Request URI: http://cspb.teletalk.com.bd/time.php]
  > HTTP chunked response
    Content-encoded entity body (gzip): 92 bytes -> 80 bytes
    File Data: 80 bytes
  > Line-based text data: text/html (1 lines)
```